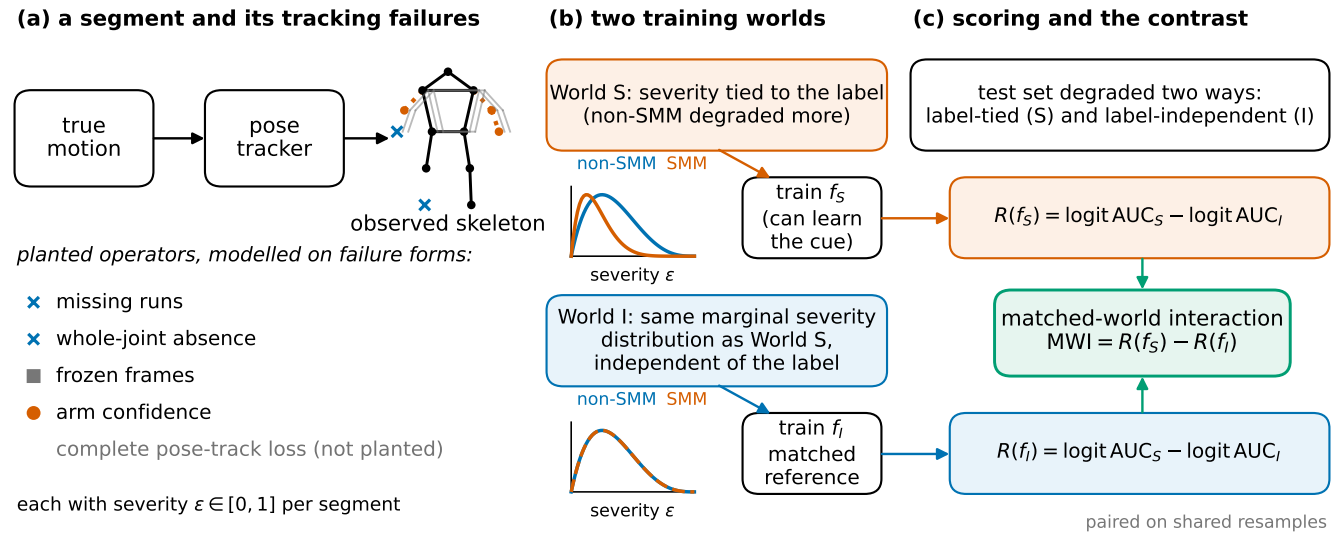
(a) a segment and its tracking failures
true motion
pose tracker
observed skeleton
planted operators, modelled on failure forms:
missing runs
whole-joint absence
frozen frames
arm confidence
complete pose-track loss (not planted)
each with severity $\varepsilon \in [0, 1]$ per segment
(b) two training worlds
World S: severity tied to the label (non-SMM degraded more)
non-SMM SMM
severity ε
train f_S (can learn the cue)
World I: same marginal severity distribution as World S, independent of the label
non-SMM SMM
severity ε
train f_I matched reference
(c) scoring and the contrast
test set degraded two ways: label-tied (S) and label-independent (I)
$R(f_S) = \text{logit AUC}_S - \text{logit AUC}_I$
matched-world interaction
$\text{MWI} = R(f_S) - R(f_I)$
$R(f_I) = \text{logit AUC}_S - \text{logit AUC}_I$
paired on shared resamples